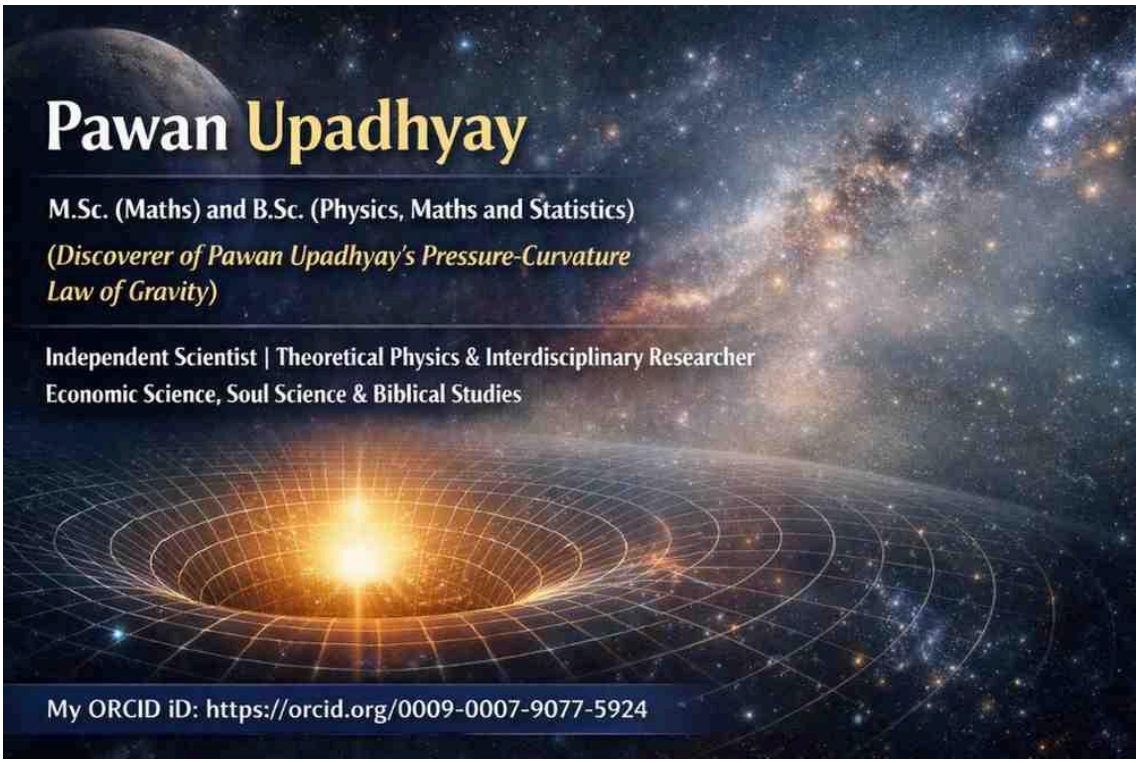
MAXIMUM GRAVITATIONAL PRESSURE AND MAXIMUM TOTAL ENERGY DENSITY NEAR BLACK HOLES COMPARED WITH NEUTRON STARS: A THEORETICAL STUDY BASED ON PAWAN UPADHYAY'S PRESSURE–CURVATURE (PPC) LAW OF GRAVITY

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Theory: Pawan Upadhyay's Pressure–Curvature Law of Gravity (PPC Law).



ABSTRACT

This research paper presents a comparative theoretical study of gravitational pressure and total energy density near neutron stars and black holes according to Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity.

The fundamental PPC equation is:

$P_g = \omega E_d$

According to the proposed PPC framework, the equation-of-state parameter ω approaches unity near both neutron stars and black holes.

However, the proposed physical relationships are different for these two gravitational environments.

Near neutron stars:

$P_g(NS) \sim E_d(NS)$

Near black holes:

$P_g(max, BH) \sim E_{max}(BH)$

The central hypothesis of this research is that maximum gravitational pressure and maximum total energy density associated with black-hole environments are greater than gravitational pressure and total energy density near neutron stars.

The proposed comparative relationships are:

$P_g(max, BH) > P_g(NS)$

$E_{max}(BH) > E_d(NS)$

And:

$\omega_{BH} > \omega_{NS}$

These relationships are presented as theoretical hypotheses requiring mathematical derivation and observational investigation.

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1. INTRODUCTION

Neutron stars and black holes represent two important classes of compact astrophysical objects associated with extreme gravitational conditions.

Neutron stars contain extremely dense matter and possess strong gravitational fields. Black holes represent highly curved spacetime regions characterized by event horizons.

Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity proposes a relationship between gravitational pressure, total energy density, and the equation-of-state parameter.

The fundamental PPC equation is:

$P_g = \omega E_d$

This research investigates the proposed differences between gravitational pressure and total energy density near neutron stars and black holes.

The central idea is that both gravitational environments approach the limiting condition $\omega \sim 1$, but the maximum gravitational pressure and maximum total energy density in the black-hole regime may exceed the corresponding quantities near neutron stars.

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2. FUNDAMENTAL EQUATIONS OF PPC LAW OF GRAVITY

The fundamental PPC equation is:

$P_g = \omega E_d$

Where:

• P_g = Gravitational pressure.

• ω (Omega) = Equation-of-state parameter.

• E_d = Total energy density.

The total energy density is defined as:

$E_d = \rho c^2$

Where:

• ρ (Rho) = Total mass-equivalent density.

• c = Speed of light in vacuum.

Combining both equations:

$P_g = \omega \rho c^2$

The equation-of-state parameter is:

$\omega = P_g / E_d$

This relationship is fundamental to the comparative analysis developed in this research paper.

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3. GRAVITATIONAL PRESSURE NEAR NEUTRON STARS

According to the proposed PPC hypothesis, gravitational pressure near neutron stars approaches total energy density.

The proposed relationship is:

$P_g(NS) \sim E_d(NS)$

The corresponding equation-of-state parameter approaches unity:

$\omega_{NS} \sim 1$

Starting with the PPC equation:

$P_g = \omega E_d$

For neutron stars:

$P_g(NS) = \omega_{NS} E_d(NS)$

When ω_{NS} approaches unity:

$P_g(NS) \sim E_d(NS)$

This represents the proposed near-unity gravitational-pressure regime near neutron stars.

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4. MAXIMUM GRAVITATIONAL PRESSURE NEAR BLACK HOLES

According to the proposed PPC hypothesis, black-hole environments represent an extreme gravitational regime in which gravitational pressure approaches its maximum limiting value.

The proposed relationship is:

$P_g(max, BH) \sim E_{max}(BH)$

The maximum limiting condition of PPC is:

$\omega \rightarrow 1$

Starting with:

$P_g = \omega E_d$

At the maximum limiting condition:

$P_g(max) = E_{max}$

Therefore, for the proposed black-hole regime:

$P_{g(max, BH)} \sim E_{max(BH)}$

This represents the proposed maximum gravitational-pressure and maximum total-energy-density relationship near black holes.

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5. MAXIMUM LIMITING CONDITION OF PPC LAW OF GRAVITY

According to the PPC framework, the maximum limiting conditions are:

$\rho \rightarrow \rho_{max}$

$E_d \rightarrow E_{max}$

$\omega \rightarrow 1$

$P_g \rightarrow P_{g(max)}$

The total maximum energy density is:

$E_{max} = \rho_{max} c^2$

At the maximum limiting condition:

$P_{g(max)} = E_{max}$

Therefore:

$P_{g(max)} = \rho_{max} c^2$

This establishes the proposed relationship between maximum gravitational pressure and maximum total energy density in the PPC framework.

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6. COMPARATIVE RELATIONSHIP BETWEEN NEUTRON STARS AND BLACK HOLES

The central hypothesis of this research is that black-hole environments can attain greater gravitational pressure and total energy density than neutron-star environments.

A. Gravitational Pressure

Near neutron stars:

$P_{g(NS)} \sim E_{d(NS)}$

Near black holes:

$P_{g(max, BH)} \sim E_{max(BH)}$

The proposed comparative relationship is:

$P_{g(max, BH)} > P_{g(NS)}$

B. Total Energy Density

Near neutron stars:

$E_{d(NS)}$

Near black holes:

$E_{max(BH)}$

The proposed comparative relationship is:

$E_{max(BH)} > E_{d(NS)}$

C. Equation-of-State Parameter

Near neutron stars:

$\omega_{NS} \sim 1$

Near black holes:

$\omega_{BH} \sim 1$

The additional proposed relationship is:

$\omega_{BH} > \omega_{NS}$

Thus, the PPC hypothesis proposes that both environments approach unity, while the equation-of-state parameter is relatively greater in the black-hole regime.

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7. MATHEMATICAL RELATIONSHIP BETWEEN GRAVITATIONAL PRESSURE, ENERGY DENSITY, AND OMEGA

The fundamental PPC equation is:

$P_g = \omega E_d$

Therefore:

$\omega = P_g / E_d$

For neutron stars:

$\omega_{NS} = P_{g(NS)} / E_{d(NS)}$

For black holes:

$\omega_{BH} = P_g(BH) / E_d(BH)$

The proposed inequality is:

$\omega_{BH} > \omega_{NS}$

For this inequality to hold mathematically:

$P_g(BH) / E_d(BH) > P_g(NS) / E_d(NS)$

This distinction is important.

Greater gravitational pressure and greater total energy density near black holes do not automatically establish a greater value of omega.

The PPC framework must demonstrate that the pressure-to-energy-density ratio is greater in the black-hole regime.

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8. PHYSICAL INTERPRETATION OF THE PPC HYPOTHESIS

A. Neutron-Star Regime

• Neutron stars contain extremely dense matter.

• Their gravitational fields are very strong.

• The PPC gravitational pressure approaches total energy density.

• The equation-of-state parameter approaches unity.

The proposed relationship is:

$P_g(NS) \sim E_d(NS)$

$\omega_{NS} \sim 1$

B. Black-Hole Regime

• Black holes are associated with extreme spacetime curvature.

• The PPC framework proposes a greater maximum total energy density than the total energy density near neutron stars.

• Maximum gravitational pressure approaches maximum total energy density.

• The equation-of-state parameter also approaches unity.

The proposed relationships are:

$P_g(max, BH) \sim E_{max}(BH)$

$\omega_{BH} \sim 1$

The proposed comparative hierarchy is:

$P_g(\max, BH) > P_g(NS)$

$E_{max}(BH) > E_d(NS)$

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9. PPC PERFECT-FLUID ENERGY-MOMENTUM TENSOR

The PPC perfect-fluid energy-momentum tensor is:

$T^{\mu\nu}_{PPC} = (E_d + P_g)u^\mu u^\nu + P_g g^{\mu\nu}$

Where:

• $T^{\mu\nu}_{PPC}$ = PPC energy-momentum tensor.

• E_d = Total energy density.

• P_g = Gravitational pressure.

• u^μ = Contravariant four-velocity.

• $g^{\mu\nu}$ = Contravariant spacetime metric.

Using the PPC equation of state:

$P_g = \omega E_d$

The tensor becomes:

$T^{\mu\nu}_{PPC} = E_d [(1 + \omega)u^\mu u^\nu + \omega g^{\mu\nu}]$

This equation provides the mathematical starting point for investigating the proposed variation of omega near neutron stars and black holes.

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10. ENERGY-MOMENTUM CONSERVATION

The PPC energy-momentum conservation equation is:

$\nabla_\mu T^{\mu\nu}_{PPC} = 0$

Contracting with the covariant four-velocity:

$u_\nu \nabla_\mu T^{\mu\nu}_{PPC} = 0$

The resulting energy-conservation equation is:

$u^\mu \nabla_\mu E_d + (E_d + P_g)\nabla_\mu u^\mu = 0$

Define fluid expansion:

$\theta = \nabla_\mu u^\mu$

Therefore:

$u^\mu \nabla_\mu E_d + (E_d + P_g)\theta = 0$

Substituting:

$P_g = \omega E_d$

We obtain:

$u^\mu \nabla_\mu E_d = -(1 + \omega)E_d \theta$

This equation describes the evolution of total energy density along the fluid flow within the PPC perfect-fluid framework.

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11. MATHEMATICAL REQUIREMENTS FOR ESTABLISHING THE PPC HYPOTHESIS

The proposed comparative relationships require further mathematical investigation.

The following requirements are important:

1. Variation of Omega

The PPC framework must derive how omega varies with total energy density and gravitational conditions.

2. Gravitational Pressure

The theory must demonstrate why maximum gravitational pressure in the black-hole regime exceeds gravitational pressure near neutron stars.

3. Total Energy Density

The theory must establish the proposed relationship:

$E_{max(BH)} > E_{d(NS)}$

4. Pressure-to-Energy-Density Ratio

The theory must independently establish:

$\omega_{BH} > \omega_{NS}$

5. Maximum Limiting Conditions

The PPC framework must specify whether maximum gravitational pressure and maximum total energy density are universal limits or environment-dependent limiting quantities.

6. Mathematical Stability and Causality

The consequences of the PPC perfect-fluid tensor must be examined for stability, causality, and energy conditions, particularly if omega approaches or exceeds unity.

7. Observational Investigation

The proposed relationships must be compared with neutron-star observations and black-hole astrophysical constraints.

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12. CONCLUSION

This research paper presents a comparative theoretical hypothesis based on Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity.

The principal distinction between the two gravitational environments is expressed through the following relationships.

Near Neutron Stars:

$P_g(NS) \sim E_d(NS)$

$\omega_{NS} \sim 1$

Near Black Holes:

$P_g(max, BH) \sim E_{max}(BH)$

$\omega_{BH} \sim 1$

The proposed comparative relationships are:

$P_g(max, BH) > P_g(NS)$

$E_{max}(BH) > E_d(NS)$

$\omega_{BH} > \omega_{NS}$

The central proposition is that neutron stars and black holes may approach the near-unity equation-of-state regime while occupying different levels of gravitational pressure and total energy density.

The PPC hypothesis proposes that the black-hole regime can attain greater maximum gravitational pressure and maximum total energy density than the neutron-star regime.

The proposed difference in omega must be established through the pressure-to-energy-density ratio and the governing PPC equations.

Further mathematical derivation, theoretical consistency analysis, and observational investigation are required to establish these propositions as physical predictions.

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13. CENTRAL SCIENTIFIC PROPOSITION

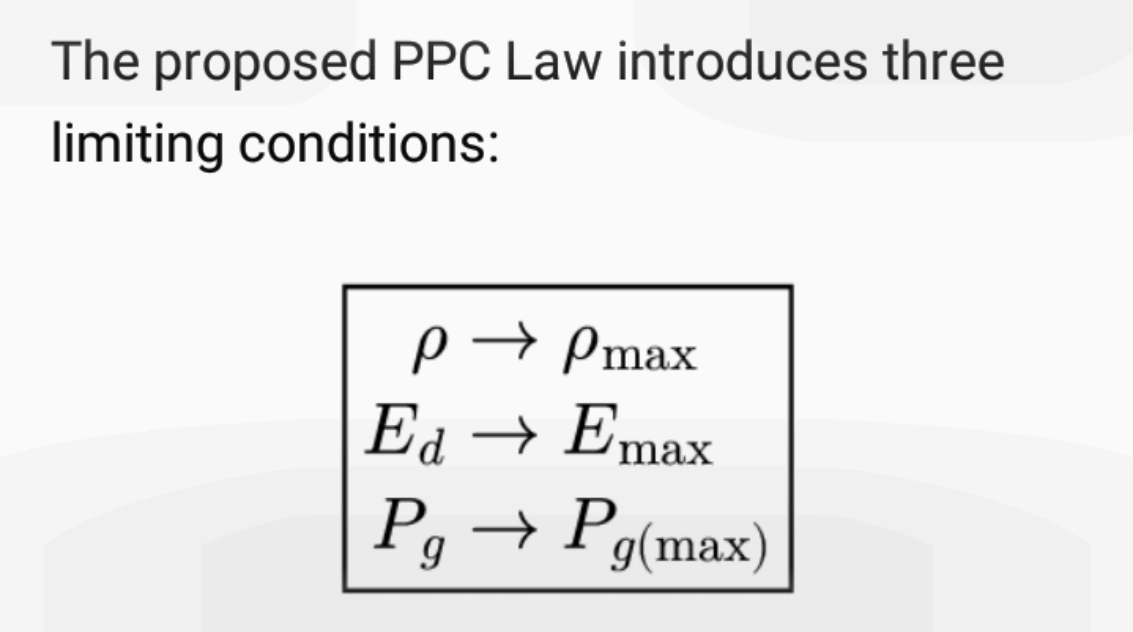
According to the proposed Pawan Upadhyay Pressure–Curvature (PPC) Law of Gravity:

Near neutron stars:

P_g(NS) ~ E_d(NS)

Near black holes:

P_g(max, BH) ~ E_max(BH)



$$\rho = \frac{dm}{dV}$$

$$E_d = \frac{dE}{dV} \rightarrow E_{\max}$$

$$\begin{aligned} \rho &\rightarrow \rho_{\max} \\ E_d &\rightarrow E_{\max} \\ \omega &\rightarrow 1 \\ P_g &\rightarrow P_{g(\max)} \end{aligned}$$

The black-hole regime is proposed to attain greater maximum gravitational pressure and maximum total energy density than the neutron-star regime.

Therefore, the proposed comparative hierarchy is:

P_g(max, BH) > P_g(NS)

E_max(BH) > E_d(NS)

With:

ω_BH > ω_NS

While:

$\omega_{BH} \sim 1$ and $\omega_{NS} \sim 1$

This is the central theoretical hypothesis investigated in this research paper.

AUTHORSHIP AND THEORY

Author: Pawan Upadhyay

Theory: Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity

Research Category: Theoretical Gravitational Physics

Research Status: Proposed theoretical hypothesis requiring mathematical derivation, consistency analysis, and observational testing.

SPECIAL NOTE: HYPOTHETICAL NUMERICAL COMPARISON OF GRAVITATIONAL PRESSURE AND TOTAL ENERGY DENSITY NEAR NEUTRON STARS AND BLACK HOLES UNDER PAWAN UPADHYAY'S PPC LAW OF GRAVITY

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Research Subject: Black Hole Physics, Pressure Curvature, Perfect Fluids, Maximum Energy Density, Maximum Mass-Equivalent Density, and Relativistic Momentum Conservation.
Theory: Pawan Upadhyay's Pressure–Curvature Law of Gravity (PPC Law).

1. PURPOSE OF THIS NUMERICAL EXAMPLE

This note presents a hypothetical numerical comparison of gravitational pressure and total energy density near neutron stars and black holes using Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity.

The fundamental PPC equation is:

$P_g = \omega E_d$

Where:

P_g = Gravitational pressure.

ω = Equation-of-state parameter.

E_d = Total energy density.

The purpose is to examine whether the proposed PPC inequalities can be satisfied using illustrative numerical values.

2. HYPOTHETICAL NUMERICAL VALUES NEAR NEUTRON STARS

Assume:

$E_d(NS) = 5.0 \times 10^{34}$ J/m³

$\omega_{NS} = 0.990$

According to the PPC equation:

$P_g(NS) = \omega_{NS} \times E_d(NS)$

Substituting:

$P_g(NS) = 0.990 \times 5.0 \times 10^{34}$

Therefore:

$P_g(NS) = 4.95 \times 10^{34}$ Pa

Thus, the hypothetical neutron-star values are:

Total energy density:

$E_d(NS) = 5.0 \times 10^{34}$ J/m³

Equation-of-state parameter:

$\omega_{NS} = 0.990$

Gravitational pressure:

$P_g(NS) = 4.95 \times 10^{34}$ Pa

3. HYPOTHETICAL NUMERICAL VALUES NEAR BLACK HOLES

Assume:

$E_{max(BH)} = 1.0 \times 10^{35}$ J/m³

$\omega_{BH} = 0.999$

According to the PPC equation:

$P_{g(max, BH)} = \omega_{BH} \times E_{max(BH)}$

Substituting:

$P_{g(max, BH)} = 0.999 \times 1.0 \times 10^{35}$

Therefore:

$P_{g(max, BH)} = 9.99 \times 10^{34}$ Pa

Thus, the hypothetical black-hole values are:

Maximum total energy density:

$E_{max(BH)} = 1.0 \times 10^{35}$ J/m³

Equation-of-state parameter:

$\omega_{BH} = 0.999$

Maximum gravitational pressure:

$P_{g(max, BH)} = 9.99 \times 10^{34}$ Pa

4. COMPARISON OF NEUTRON STARS AND BLACK HOLES

The hypothetical numerical comparison is:

NEUTRON STARS:

$E_{d(NS)} = 5.0 \times 10^{34}$ J/m³

$P_{g(NS)} = 4.95 \times 10^{34}$ Pa

$\omega_{NS} = 0.990$

BLACK HOLES:

$E_{max(BH)} = 1.0 \times 10^{35}$ J/m³

$P_{g(max, BH)} = 9.99 \times 10^{34}$ Pa

$\omega_{BH} = 0.999$

5. VERIFICATION OF THE PROPOSED PPC INEQUALITIES

A. Gravitational pressure comparison:

$9.99 \times 10^{34} > 4.95 \times 10^{34}$

Therefore:

$P_g(\max, BH) > P_g(NS)$

B. Total energy density comparison:

$1.0 \times 10^{35} > 5.0 \times 10^{34}$

Therefore:

$E_{max}(BH) > E_d(NS)$

C. Equation-of-state parameter comparison:

$0.999 > 0.990$

Therefore:

$\omega_{BH} > \omega_{NS}$

D. Near-unity limiting condition:

$\omega_{NS} = 0.990$, approaching unity.

$\omega_{BH} = 0.999$, approaching unity.

Both values approach unity, while the hypothetical black-hole value is greater than the neutron-star value.

6. MAXIMUM LIMITING RELATIONSHIP

According to the PPC Law of Gravity:

$P_g = \omega E_d$

At the maximum limiting condition:

$\omega \rightarrow 1$

Therefore:

$P_{g(max)} = E_{max}$

For the hypothetical black-hole example:

$P_{g(max, BH)} = 9.99 \times 10^{34}$ Pa

$E_{max(BH)} = 1.0 \times 10^{35}$ J/m³

Thus:

$P_{g(max, BH)} \sim E_{max(BH)}$

This illustrates the proposed PPC relationship between maximum gravitational pressure and maximum total energy density.

7. IMPORTANT MATHEMATICAL OBSERVATION

The PPC equation can be rearranged as:

$\omega = P_g / E_d$

Therefore, greater gravitational pressure and greater total energy density do not automatically establish a greater value of omega.

The proposed inequality:

$\omega_{BH} > \omega_{NS}$

must be verified independently through the pressure-to-energy-density ratio.

The numerical example satisfies all three proposed inequalities because the assumed values have been selected accordingly.

8. SCIENTIFIC QUALIFICATION

All numerical values in this note are hypothetical examples selected for mathematical illustration.

They are not experimentally measured values of gravitational pressure, total energy density, or omega near actual neutron stars or black holes.

The calculations demonstrate mathematical compatibility of the proposed PPC relationships, not their observational confirmation.

Further mathematical derivation from the PPC gravitational field equations, physical consistency analysis, and comparison with astrophysical observations are necessary.

9. CONCLUSION

This numerical example illustrates the proposed comparative hierarchy in Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity.

The hypothetical relationships are:

$P_{g(max, BH)} > P_{g(NS)}$

$E_{max(BH)} > E_{d(NS)}$

$\omega_{BH} > \omega_{NS}$

With:

$\omega_{BH} \sim 1$

$\omega_{NS} \sim 1$

The example demonstrates that both neutron-star and black-hole regimes can approach unity while the black-hole regime has greater hypothetical gravitational pressure and total energy density.

CENTRAL NUMERICAL PROPOSITION

Near neutron stars:

$P_{g(NS)} = 4.95 \times 10^{34}$ Pa

$E_{d(NS)} = 5.0 \times 10^{34}$ J/m³

$\omega_{NS} = 0.990$

Near black holes:

$P_{g(max, BH)} = 9.99 \times 10^{34}$ Pa

$E_{max(BH)} = 1.0 \times 10^{35}$ J/m³

$\omega_{BH} = 0.999$

These values provide an illustrative numerical example for further mathematical investigation of the PPC Law of Gravity.

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Theory: Pawan Upadhyay's Pressure–Curvature (PPC) Law of Gravity